Good riddance: Thymocyte clonal deletion prevents autoimmunity.

Clonal deletion is arguably the most important mechanism of eliminating self-reactive thymocytes from the T-cell repertoire. Recent work has identified new players in this process. On the thymocyte side, several molecules have been newly implicated in the pathway from initial T-cell receptor signaling through to the final result: gene transcription and thymocyte apoptosis. In addition, several proapoptotic molecules have been found to be necessary for the death of self-reactive thymocytes. On the antigen-presenting cell side, the expression of peripheral self-antigens, regulated at least in part by the autoimmune regulator (AIRE) protein, is crucial for complete elimination of autoreactive thymocytes. The importance of thymic peripheral antigen expression and clonal deletion to self-tolerance is demonstrated in the autoimmune diseases autoimmune-polyendocrinopathy-candidiasis-ectodermal dystrophy and type-1 diabetes mellitus.